

OTT Platform Case Study Assignment – MongoDB Replication & Replica Sets

Case Study Title

Designing a Scalable OTT Streaming Analytics Platform using MongoDB Replica Sets

Scenario

An OTT streaming company similar to Netflix or Disney+ wants to build a backend system to manage:

- User streaming history
- Personalized recommendations
- Watch activity logs
- Device usage tracking
- Trending content analytics

The platform serves millions of users daily. Since users watch content continuously, the company requires:

- High availability
- Fault tolerance
- Data redundancy
- Read scalability
- Continuous monitoring

To achieve this, the engineering team decides to implement **MongoDB Replication using Replica Sets**.

Assignment Objective

Students must design and implement a beginner-friendly MongoDB-based OTT analytics system that demonstrates:

- Replica Set Architecture
- Primary-Secondary Replication
- High Availability Concepts
- Replication Monitoring

The assignment focuses on understanding **MongoDB replication concepts practically** through an OTT use case.

Business Requirements

The OTT platform should support the following modules:

```
mongodb-replication-ott/  
├── data/  
│   ├── db1/  
│   ├── db2/  
│   └── db3/  
└── scripts/  
    ├── setupReplica.js  
    ├── insertUsers.js  
    ├── insertStreamingHistory.js  
    ├── insertRecommendations.js  
    ├── insertActivityLogs.js  
    ├── checkData.js  
    ├── updateData.js  
    ├── deleteData.js  
    └── failoverDemo.js
```

Purpose of Each Folder

Folder/File	Purpose
data/db1	Primary node database files
data/db2	Secondary node database files
data/db3	Secondary node database files
scripts	MongoDB operation scripts

Collections and Columns

1. users Collection

users

Column	Type	Description
--------	------	-------------

userId	String	Unique user ID
name	String	User name
email	String	Email address
subscription	String	Basic/Premium
country	String	User country
createdAt	Date	Account creation date

2. streaming_history Collection

streaming_history

Column	Type	Description
historyId	String	Watch history ID
userId	String	Linked user
contentTitle	String	Movie/Series name
genre	String	Content genre
watchDuration	Number	Minutes watched
watchedAt	Date	Watch timestamp

3. recommendations Collection

recommendations

Column	Type	Description
recommendationId	String	Recommendation ID
userId	String	Linked user
recommendedTitle	String	Suggested content
genre	String	Recommendation genre

score	Number	Recommendation score
-------	--------	----------------------

CRUD Operations Requirements

CREATE Operations

Students should create scripts for:

Script	Operation
insertUsers.js	Insert OTT users
insertStreamingHistory.js	Insert watch history
insertRecommendations.js	Insert recommendations

READ Operations

Using:

`checkData.js`

Students should:

- View all users
- View streaming history
- View recommendations
- View logs

UPDATE Operations

Using:

`updateData.js`

Students should:

- Update subscription plan
- Update watch duration
- Update recommendation score

DELETE Operations

Using:

`deleteData.js`

Students should:

- Delete inactive users
- Delete old activity logs
- Delete watch history

Replica Set Operations

`setupReplica.js`

Students should:

- Initialize replica set
- Configure db1, db2, db3
- Set Primary and Secondary nodes

`failoverDemo.js`

Students should:

- Stop Primary node
- Observe automatic election
- Verify new Primary node

Beginner Workflow

Step 1 — Start MongoDB Nodes

Run:

- db1 on port 27017
- db2 on port 27018
- db3 on port 27019

Step 2 — Configure Replica Set

Run:

`setupReplica.js`

Step 3 — Insert OTT Data

Run:

```
insertUsers.js  
insertStreamingHistory.js  
insertRecommendations.js  
insertActivityLogs.js
```

Step 5 — Perform CRUD Operations

Run:

```
checkData.js  
updateData.js  
deleteData.js
```